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Sensor-Based Nadi Pariksha Diagnostic Machine

A Digital Approach to Ayurvedic Pulse Diagnosis

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Ayurveda & Nadi Pariksha: Technical Background

Ayurveda explains physiological state through Tridosha theory

- Vata → movement, rhythm
- Pitta → metabolism, heat
- Kapha → structure, stability

Nadi Pariksha is performed at the radial artery

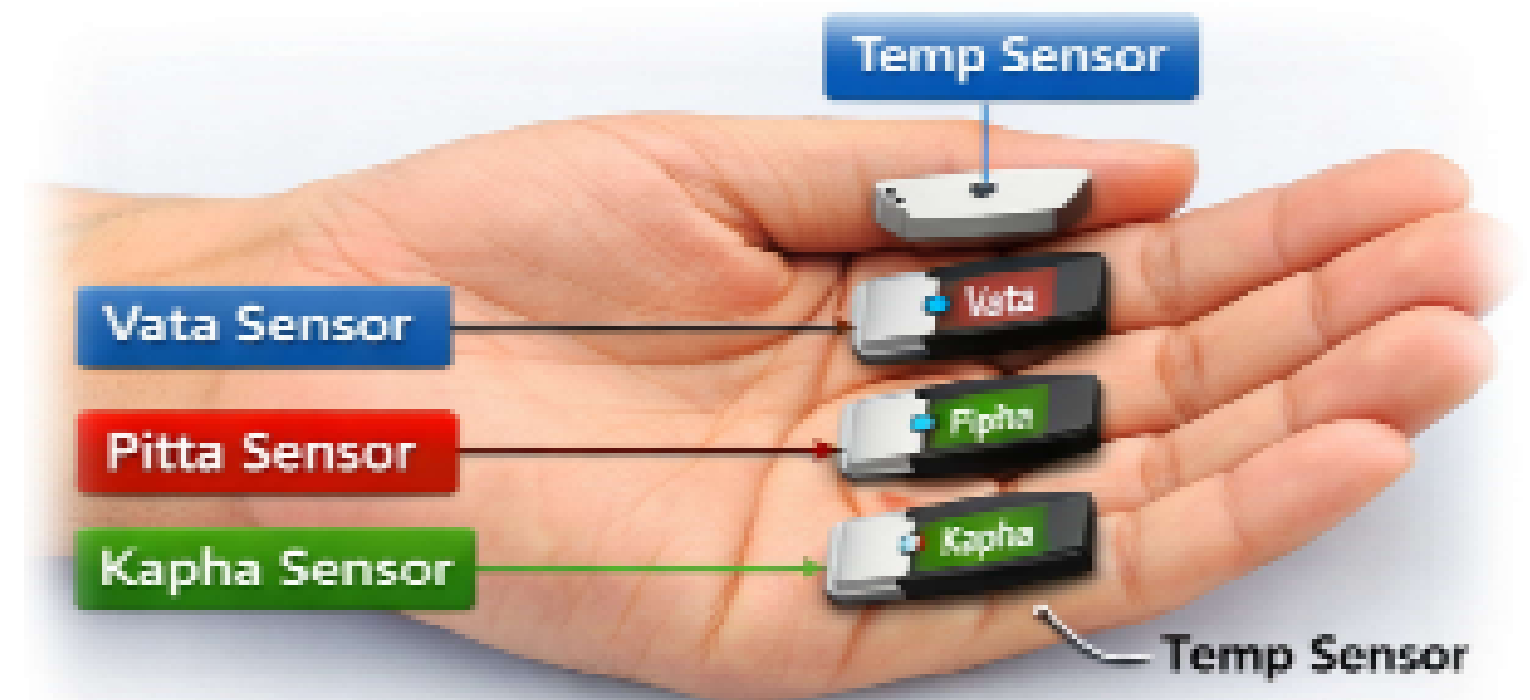
Three finger placements correspond to:

- Index → Vata
- Middle → Pitta
- Ring → Kapha

Pulse attributes analyzed:

- Rate, rhythm, strength, temperature, waveform behavior

Traditionally qualitative → needs digitization



Limitations of Traditional Diagnosis

1. Highly dependent on:

- Practitioner experience
- Finger sensitivity
- Time of examination

2. Scientific challenges:

- Low inter-rater agreement
- No numeric thresholds
- No electronic storage

3. Operational limitations:

- Not scalable
- Difficult for telemedicine

4. Research gap:

- Lack of standardized sensor-based Nadi systems validated scientifically

Research Objectives & Design Goals

Primary Objectives

- Digitize pulse acquisition at Vata–Pitta–Kapha points
- Quantify Ayurvedic pulse parameters
- Reduce observer bias
- Enable ML-supported interpretation

Design Goals

- Non-invasive and low-cost
- Easy sensor placement
- Real-time visualization
- Self-setup admin interface
- IEEE-compatible research output

Complete System Architecture

Hardware Subsystem

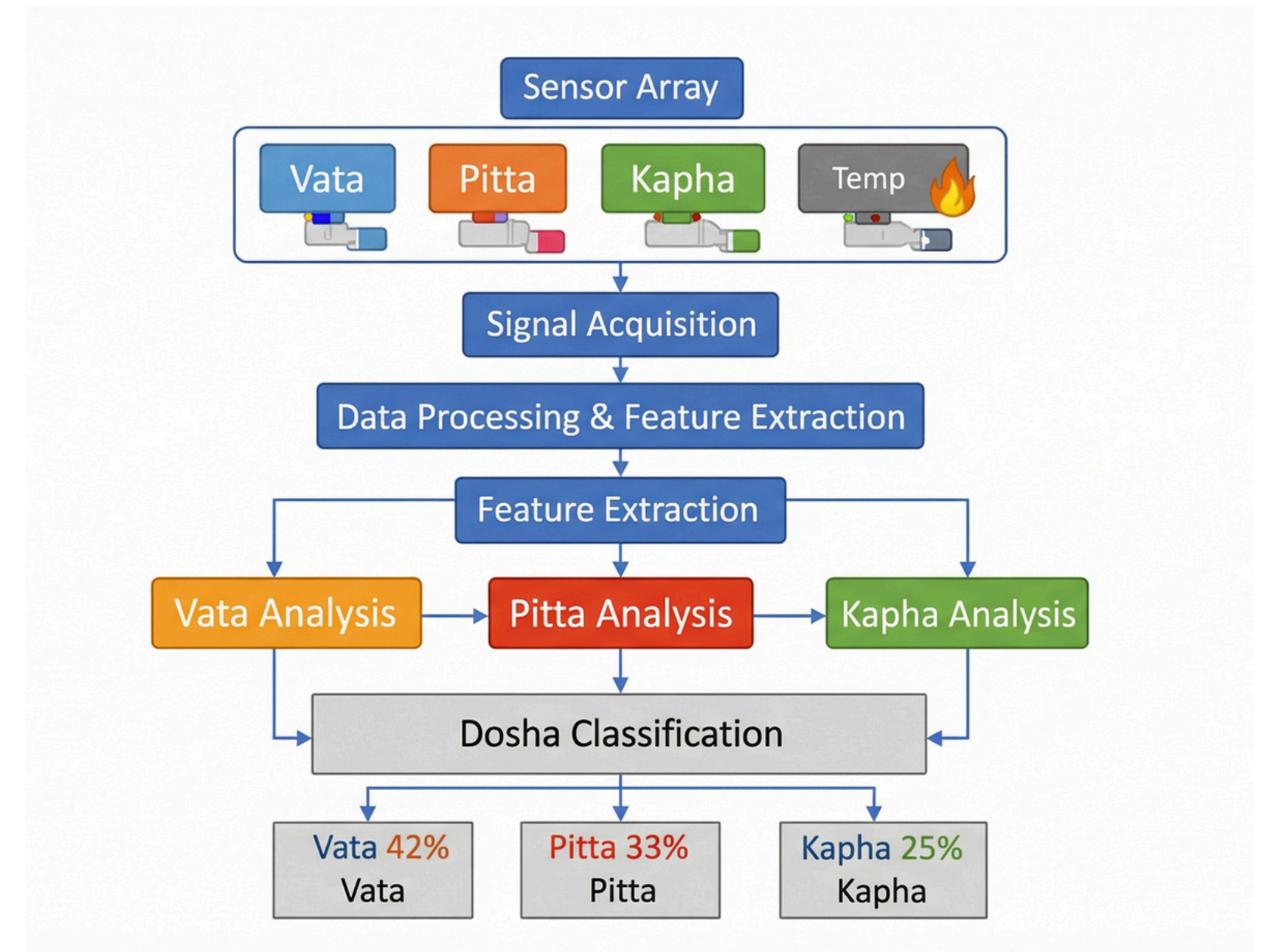
- Pressure sensors: arterial pulse force
- PPG sensors: blood volume variation
- Temperature sensor: skin-level Tapamana

Embedded Subsystem

- Microcontroller with multi-channel ADC
- Signal sampling (100–500 Hz)
- Noise filtering & artifact removal

Software Subsystem

- Feature extraction engine
- Rule-based Ayurvedic logic
- ML-based dosha prediction
- Disease risk estimation
- Web/mobile dashboard



Dosha Mapping & ML Prediction

1. Two-layer interpretation:

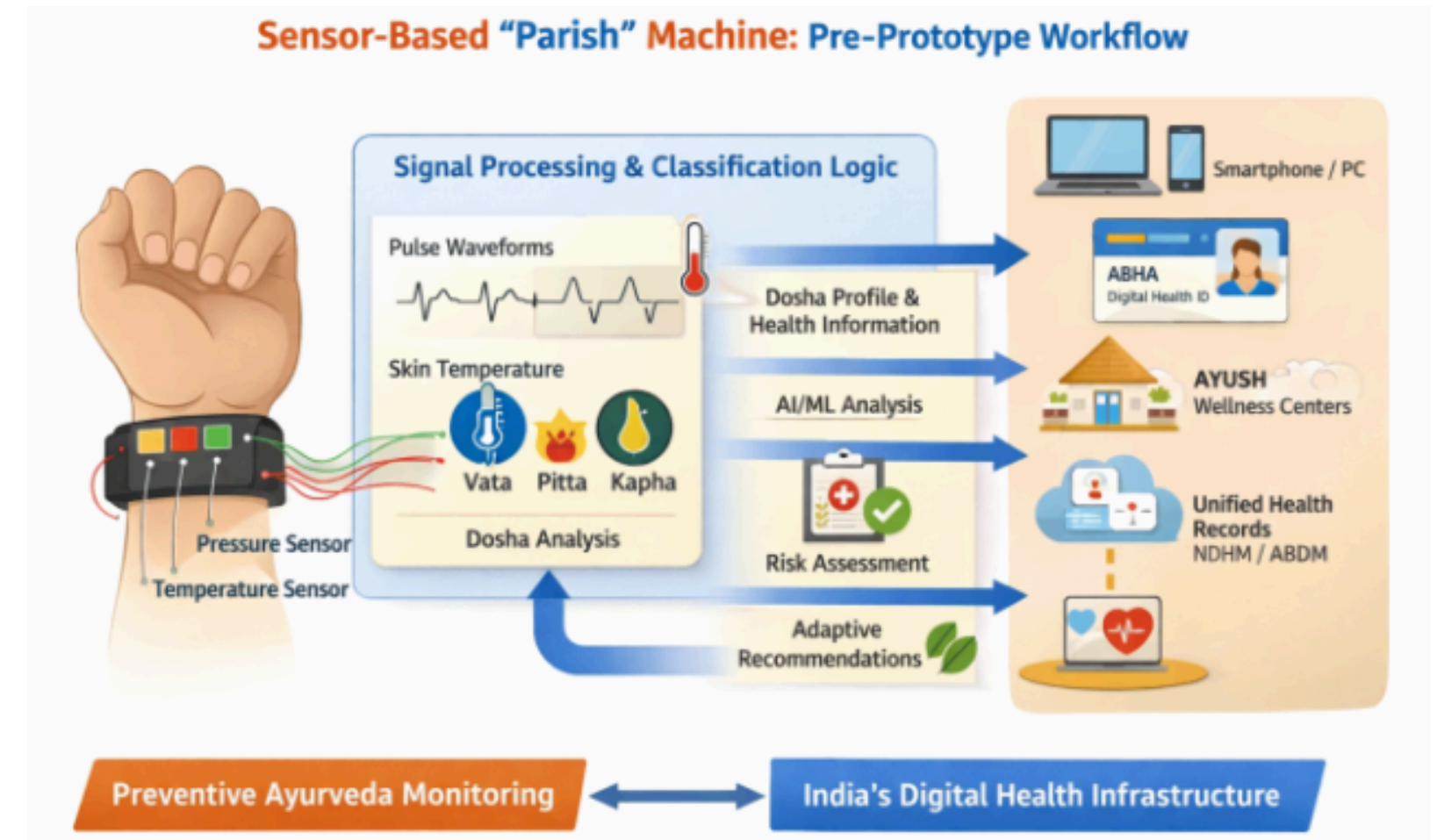
- Layer 1: Ayurvedic Rules
Classical pulse–dosha correlations
- Layer 2: Machine Learning
Supervised learning using expert-labeled data

2. Input Features:

- Pulse rate
- RR variability
- Amplitude
- Temperature

3. Outputs:

- Dosha dominance percentages
- Confidence score for prediction reliability



Disease Risk Assessment & Confidence Modeling

Disease risk is not diagnostic but preventive:

1. Risk derived from:

- Dosha imbalance magnitude
- Persistent abnormal pulse patterns

2. Risk Levels:

- Low → Balanced dosha
- Medium → Mild imbalance
- High → Persistent deviation

3. Confidence Score:

- Based on signal quality
- ML prediction probability
- Consistency across sensors

Dashboard & Admin Self-Setup System

User Dashboard

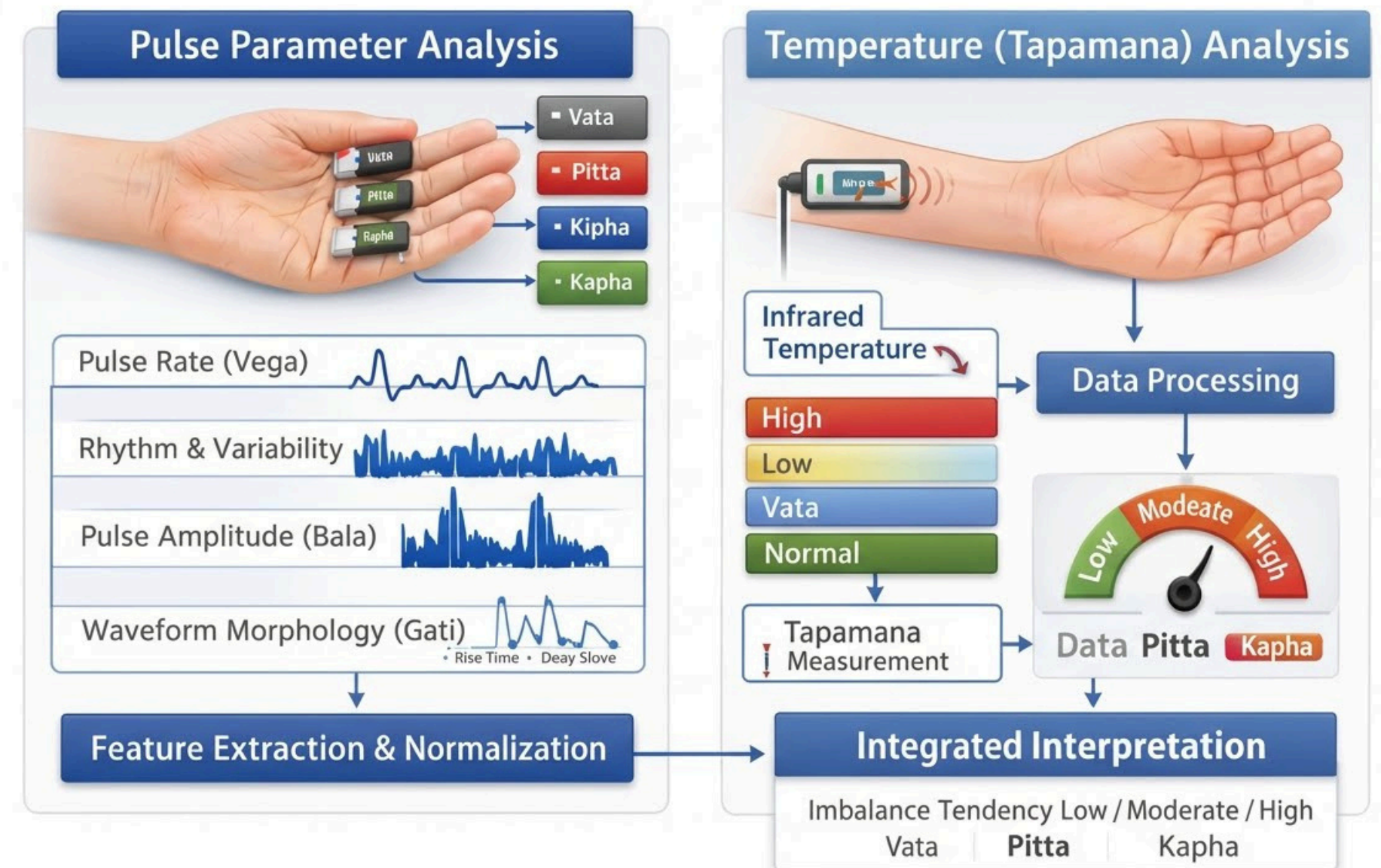
- Real-time waveforms
- Parameter tiles
- Dosha charts
- ML prediction & confidence display

Admin Dashboard

- Guided sensor connection wizard
- Visual wrist placement guide
- Automatic sensor detection
- Calibration & validation tools
- ML model control panel

Key Innovation

- Eliminates need for Ayurvedic or technical experts during setup



Validation, Impact & Future Scope

1. Validation Plan

- Double-blind comparison with expert practitioners
- Agreement analysis (Cohen's Kappa)
- Iterative ML refinement

2. Impact

- Preventive healthcare screening
- Digital Ayurveda integration
- Rural & telemedicine deployment

3. Future Scope

- Clinical trials
- Large-scale data collection
- Advanced AI models
- Integration with national digital health platforms

Thank You!

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